Certificate of Calibration
Report Number: 778302

Sensor Model: CX-1050-SD-HT-1.4L	Serial Number: X93303
Sensor Type: Cernox Resistor	Sales Order: 90141-1
Sensor Excitation: see *Test Data* page of report	Date: April 09, 2014
Temperature Range: 1.40K to 325K	Due: April 09, 2015

Traceability and Calibration Method

This temperature sensor has been calibrated to the International Temperature Scale of 1990 (ITS-90) or the Provisional Low Temperature Scale (PLTS-2000) as appropriate. The calibrations are traceable to the National Institute of Standards and Technology (NIST, United States), the National Physical Laboratory (NPL, United Kingdom), the Physikalisch-Technische Bundesanstalt (PTB, Germany), or natural physical constants.

Lake Shore Cryotronics maintains ITS-90 and PLTS-2000 on standard platinum (PRT), rhodium-iron (RIRT), and germanium (GRT) resistance thermometers that have been calibrated directly by an internationally recognized national metrology institute (NIST, NPL, PTB) for T < 330 K or an ISO 17025 accredited metrology laboratory for 330 K < T < 800 K. A nuclear orientation thermometer is also used for temperatures less than 50 mK. These standards are routinely intercompared to verify consistency and accuracy of the temperature scale.

The sensor calibrations are performed by comparison to laboratory standard resistance thermometers and tested in accordance with Lake Shore Cryotronics, Inc. Quality Assurance Manual (QP-4220). The quality system of Lake Shore Cryotronics is registered to ISO 9001:2008.

Procedures used: 021-97-02, 099-00-00, 121-96-02, 029-95-02

Notes

The calibration results in this report apply only to the specific sensor specified above.

Unless stated otherwise, the uncertainties in this report are based on an approximate 95% confidence level with a coverage factor k=2.

Reported by: Todd Rittershausen
Calibration Engineer/Technician

Approved by: John Krause
Metrology

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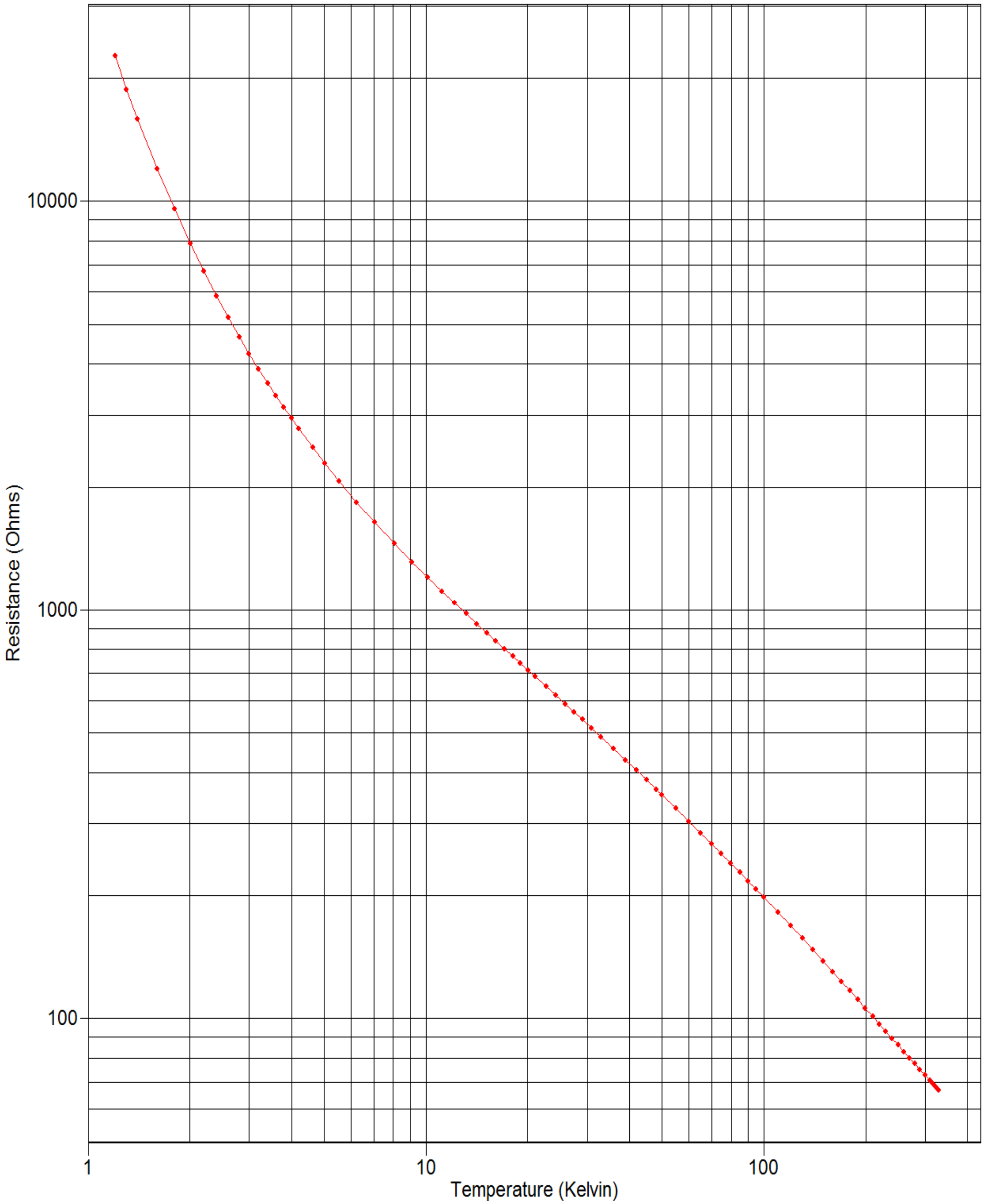
DATA PLOT

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K



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TEST DATA

Calibration Report: 778302
Sales Order: 90141-1
Sensor Model: CX-1050-SD-HT-1.4L
Serial Number: X93303
Sensor Type: Cernox Resistor
Temperature Range: 1.40K to 325K

Index	Temp. (K)	Resistance (Ω)	Excitation	Index	Temp. (K)	Resistance (Ω)	Excitation
1	1.20156	22724.4	2mV±25%	46	41.9797	404.986	2mV±25%
2	1.30197	18764.3	2mV±25%	47	44.9760	383.397	2mV±25%
3	1.40298	15856.0	2mV±25%	48	47.9825	364.132	2mV±25%
4	1.60187	12003.2	2mV±25%	49	49.9777	352.430	2mV±25%
5	1.80143	9563.73	2mV±25%	50	54.9859	326.268	2mV±25%
6	2.00611	7880.16	2mV±25%	51	59.9853	303.954	2mV±25%
7	2.20126	6739.71	2mV±25%	52	64.9802	284.636	2mV±25%
8	2.40237	5862.78	2mV±25%	53	69.9898	267.693	2mV±25%
9	2.60330	5189.49	2mV±25%	54	74.9865	252.750	2mV±25%
10	2.80208	4664.09	2mV±25%	55	79.9819	239.444	2mV±25%
11	3.00040	4239.55	2mV±25%	56	84.9800	227.480	2mV±25%
12	3.20110	3885.98	2mV±25%	57	89.9802	216.687	2mV±25%
13	3.40203	3590.90	2mV±25%	58	94.9778	206.925	2mV±25%
14	3.60168	3341.98	2mV±25%	59	99.9797	197.979	2mV±25%
15	3.80413	3126.43	2mV±25%	60	109.986	182.236	2mV±25%
16	3.99674	2947.68	2mV±25%	61	119.978	168.811	2mV±25%
17	4.20886	2776.30	2mV±25%	62	129.977	157.223	2mV±25%
18	4.62919	2497.47	2mV±25%	63	139.984	147.086	2mV±25%
19	5.03714	2280.77	2mV±25%	64	149.986	138.174	2mV±25%
20	5.53582	2069.40	2mV±25%	65	159.985	130.271	2mV±25%
21	6.24952	1837.20	2mV±25%	66	169.982	123.210	2mV±25%
22	7.06357	1638.49	2mV±25%	67	179.982	116.871	2mV±25%
23	8.08338	1452.79	2mV±25%	68	189.982	111.151	2mV±25%
24	9.10572	1312.41	2mV±25%	69	199.982	105.969	2mV±25%
25	10.1265	1202.23	2mV±25%	70	209.980	101.262	2mV±25%
26	11.1452	1113.16	2mV±25%	71	219.973	96.9594	2mV±25%
27	12.1535	1039.77	2mV±25%	72	229.981	93.0149	2mV±25%
28	13.1549	977.862	2mV±25%	73	239.988	89.3858	2mV±25%
29	14.1503	924.783	2mV±25%	74	249.985	86.0521	2mV±25%
30	15.1394	878.612	2mV±25%	75	259.972	82.9729	2mV±25%
31	16.1269	837.746	2mV±25%	76	269.987	80.1097	2mV±25%
32	17.1064	801.582	2mV±25%	77	279.991	77.4566	2mV±25%
33	18.0856	768.901	2mV±25%	78	289.980	74.9972	2mV±25%
34	19.0641	739.247	2mV±25%	79	299.998	72.6917	2mV±25%
35	20.0443	712.059	2mV±25%	80	310.001	70.5488	2mV±25%
36	21.1255	684.742	2mV±25%	81	315.011	69.5204	2mV±25%
37	22.7118	648.702	2mV±25%	82	320.010	68.5338	2mV±25%
38	24.3018	616.599	2mV±25%	83	326.005	67.3917	2mV±25%
39	25.8890	587.958	2mV±25%	84	329.994	66.6571	2mV±25%
40	27.4859	561.966	2mV±25%				
41	29.0811	538.398	2mV±25%				
42	30.8898	514.142	2mV±25%				
43	32.9858	488.869	2mV±25%				
44	35.9845	456.991	2mV±25%				
45	38.9774	429.335	2mV±25%				

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Calibration Data Uncertainty

The uncertainties of the measured calibration data for Lake Shore's sensors are summarized in the table below. The values given are the combined uncertainty of the temperature measurement and the resistance or voltage measurement expressed as an equivalent temperature uncertainty in millikelvin (mK). Note that the values are the calibration uncertainty only and do not include the stability of the temperature sensor. The uncertainty analysis has followed the guidelines for determining measurement uncertainty as outlined in the ISO Guide to the Expression of Uncertainty in Measurement, NIST Technical Note 1297, and ANSI/NCSL Z540-2-1997. Since the uncertainty varies with temperature due to the variation of the sensor sensitivity and excitation, the table gives typical values at several different temperatures throughout the range of the calibration. The uncertainty is based on an approximate 95% confidence level with a coverage factor k = 2.

T (K)	Uncertainty (± mK)												
	GR	Cernox (CX)					RX			Platinum		RF-800	Diode
		1010	1030	1050	1070	1080	102A	103A	202A	100 Ω	25 Ω	27 Ω	
1.4	4	4	4	4			4	4	4			5	7
4.2	4	4	4	4	4		4	6	5			5	5
10	4	5	5	4	4		10	15	12			7	6
20	8	10	9	8	8	8	35	35	28	9	10	13	9
30	9	13	11	9	9	9	76	61	46	9	9	14	31
50	11	18	14	12	12	11				10	10	13	37
100	20	29	22	17	16	14				11	12	12	32
300		78	60	46	45	36				24	24	25	35
400		124	94	74	72	60				45	45	45	49
500										51	51		54

Polynomial Fit Uncertainty

When a sensor is used to measure temperature, a polynomial fit to the measured calibration data is often used to convert the sensor resistance (R) or voltage (V) to a temperature (T). How well the polynomial represents the sensor calibration data is another source of uncertainty when using the sensor. In the polynomials provided with this set of calibration data, the standard deviation of the fit can be used as an estimate of this additional temperature uncertainty. The standard deviation of fit is determined from the following equation:

$$\sigma_{fit}{}^2 = \frac{\sum_{i=1}^{N}(T_i - T_{icalc})^2}{N-n} = \frac{N}{N-n}(\Delta T_{RMS})^2$$

where

- σ_{fit} = standard deviation of the fit
- T_i = measured temperature for point i
- T_{icalc} = the temperature calculated from the polynomial equation for point i
- N = number of data points in fit range
- n = number of fit coefficients
- ΔT_{RMS} = root mean square deviation of fit

A value of ΔT_{RMS} is given for each range of fit.

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POLYNOMIAL EQUATION

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Polynomial Type: Chebychev
Useful Range of Fit:

1.40 K	to	14.1 K
1.590e+4 Ohms	to	924.8 Ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:
ZL = 2.92311226922 ZU = 4.35649318298

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	5.459379	2.3872E-04	22869.26
1	-6.297939	3.8325E-04	-16432.83
2	2.848206	3.3731E-04	8443.96
3	-1.077153	3.4622E-04	-3111.21
4	0.343990	3.2717E-04	1051.40
5	-0.088520	3.0121E-04	-293.88
6	0.015368	2.9954E-04	51.31
7	0.000137	3.0194E-04	0.45
8	-0.002218	3.0446E-04	-7.28

Z = Log(Resistance)

k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)

Temp. (K) = ΣA_i* COS(i * ARCCOS(k)), where 0 <= i <= 8
and the A_i's are the coefficients in the table above.

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POLYNOMIAL EQUATION

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Polynomial Type: Chebychev
Temp. (K) vs. Log(Resistance)

	R Meas. (Ω)	T Meas. (K)	T Eq. (K)	T diff. (mK)
1	22724.44	1.20156	1.20125	0.31
2	18764.28	1.30197	1.30319	-1.22
3	15856.00	1.40298	1.40192	1.06
4	12003.19	1.60187	1.60102	0.85
5	9563.731	1.80143	1.80233	-0.91
6	7880.160	2.00611	2.00739	-1.28
7	6739.708	2.20126	2.20152	-0.26
8	5862.783	2.40237	2.40168	0.69
9	5189.489	2.60330	2.60229	1.01
10	4664.087	2.80208	2.80094	1.14
11	4239.550	3.00040	2.99973	0.67
12	3885.982	3.20110	3.20107	0.03
13	3590.897	3.40203	3.40230	-0.27
14	3341.983	3.60168	3.60263	-0.95
15	3126.427	3.80413	3.80498	-0.86
16	2947.679	3.99674	3.99821	-1.47
17	2776.302	4.20886	4.21023	-1.37
18	2497.467	4.62919	4.62789	1.30
19	2280.766	5.03714	5.03651	0.63
20	2069.397	5.53582	5.53535	0.46
21	1837.202	6.24952	6.24825	1.27
22	1638.489	7.06357	7.06251	1.06
23	1452.793	8.08338	8.08412	-0.74
24	1312.406	9.10572	9.10682	-1.10
25	1202.229	10.12650	10.12774	-1.24
26	1113.157	11.14516	11.14530	-0.14
27	1039.771	12.15353	12.15313	0.40
28	977.8620	13.15493	13.15457	0.36
29	924.7828	14.15033	14.14939	0.94
30	878.6124	15.13941	15.13848	0.92
31	837.7458	16.12688	16.12820	-1.32

Order of Fit = 8 RMS error of fit = 0.93 mK
Largest absolute error = -1.47 mK at data point no. 16

POLYNOMIAL EQUATION

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Polynomial Type: Chebychev
Useful Range of Fit:

14.2 K	to	80.0 K
924.8 Ohms	to	239.4 Ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:
ZL = 2.33583305074 ZU = 3.01693787225

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	42.553899	3.6715E-04	115901.95
1	-37.855514	5.9926E-04	-63170.36
2	8.406373	5.4652E-04	15381.65
3	-1.054057	5.1363E-04	-2052.18
4	0.112147	4.9137E-04	228.24
5	-0.003961	4.6203E-04	-8.57
6	-0.007072	4.6296E-04	-15.28

Z = Log(Resistance)

k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)

Temp. (K) = ΣA_i* COS(i * ARCCOS(k)), where 0 <= i <= 6
and the A_i's are the coefficients in the table above.

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POLYNOMIAL EQUATION

Calibration Report: 778302
Sales Order: 90141-1
Sensor Model: CX-1050-SD-HT-1.4L
Serial Number: X93303
Sensor Type: Cernox Resistor
Temperature Range: 1.40K to 325K

Polynomial Type: Chebychev
Temp. (K) vs. Log(Resistance)

	R Meas. (Ω)	T Meas. (K)	T Eq. (K)	T diff. (mK)
27	1039.771	12.15313	12.15181	1.32
28	977.8620	13.15457	13.15631	-1.75
29	924.7828	14.14939	14.15111	-1.73
30	878.6124	15.13941	15.13897	0.44
31	837.7458	16.12688	16.12658	0.30
32	801.5817	17.10643	17.10416	2.27
33	768.9006	18.08557	18.08395	1.62
34	739.2465	19.06407	19.06316	0.91
35	712.0587	20.04428	20.04607	-1.79
36	684.7420	21.12553	21.12592	-0.39
37	648.7017	22.71176	22.71214	-0.38
38	616.5989	24.30176	24.30398	-2.22
39	587.9577	25.88904	25.89005	-1.02
40	561.9662	27.48591	27.48586	0.05
41	538.3977	29.08106	29.08050	0.55
42	514.1419	30.88979	30.88961	0.18
43	488.8686	32.98575	32.98344	2.31
44	456.9915	35.98447	35.98375	0.72
45	429.3351	38.97737	38.97752	-0.16
46	404.9865	41.97972	41.97794	1.78
47	383.3969	44.97602	44.98071	-4.69
48	364.1318	47.98252	47.98212	0.40
49	352.4301	49.97765	49.97622	1.43
50	326.2684	54.98587	54.98639	-0.52
51	303.9536	59.98527	59.98485	0.42
52	284.6363	64.98019	64.98157	-1.39
53	267.6933	69.98980	69.98844	1.36
54	252.7499	74.98652	74.98624	0.28
55	239.4443	79.98190	79.98023	1.67
56	227.4800	84.98003	84.98328	-3.25
57	216.6871	89.98015	89.97888	1.27

Order of Fit = 6   RMS error of fit = 1.60 mK
Largest absolute error = -4.69 mK at data point no. 47

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POLYNOMIAL EQUATION

Calibration Report: 778302
Sales Order: 90141-1
Sensor Model: CX-1050-SD-HT-1.4L
Serial Number: X93303
Sensor Type: Cernox Resistor
Temperature Range: 1.40K to 325K

Polynomial Type: Chebychev
Useful Range of Fit:

80.0 K	to	325. K
239.4 Ohms	to	67.58 Ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:
ZL = 1.82384622589 ZU = 2.42763752697

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	177.276724	1.2964E-03	136747.25
1	-126.914055	2.0022E-03	-63388.02
2	22.155536	1.9229E-03	11521.90
3	-2.983506	1.8274E-03	-1632.65
4	0.543432	1.7403E-03	312.27
5	-0.098366	1.7439E-03	-56.40
6	0.012450	1.7235E-03	7.22
7	-0.003066	1.6643E-03	-1.84

Z = Log(Resistance)

k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)

Temp. (K) = ΣA_i* COS(i * ARCCOS(k)), where 0 <= i <= 7
and the A_i's are the coefficients in the table above.

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POLYNOMIAL EQUATION

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Polynomial Type: Chebychev
Temp. (K) vs. Log(Resistance)

	R Meas. (Ω)	T Meas. (K)	T Eq. (K)	T diff. (mK)
53	267.6933	69.98844	69.98915	-0.71
54	252.7499	74.98624	74.98470	1.54
55	239.4443	79.98023	79.97784	2.38
56	227.4800	84.98003	84.98378	-3.75
57	216.6871	89.98015	89.98714	-6.99
58	206.9246	94.97779	94.97161	6.17
59	197.9790	99.97973	99.97629	3.44
60	182.2361	109.98569	109.98271	2.97
61	168.8107	119.97798	119.98472	-6.75
62	157.2230	129.97710	129.97591	1.19
63	147.0859	139.98447	139.98841	-3.93
64	138.1740	149.98557	149.98538	0.18
65	130.2710	159.98524	159.97874	6.50
66	123.2105	169.98203	169.97938	2.65
67	116.8714	179.98186	179.98133	0.53
68	111.1508	189.98213	189.98683	-4.71
69	105.9693	199.98166	199.98831	-6.65
70	101.2617	209.97982	209.97585	3.97
71	96.95937	219.97284	219.97206	0.78
72	93.01495	229.98081	229.97628	4.53
73	89.38576	239.98799	239.99463	-6.65
74	86.05213	249.98485	249.98181	3.04
75	82.97289	259.97152	259.96598	5.54
76	80.10965	269.98661	269.99077	-4.16
77	77.45665	279.99062	279.99887	-8.26
78	74.99721	289.98028	289.97197	8.31
79	72.69174	299.99815	300.00155	-3.39
80	70.54883	310.00057	309.98396	16.61
81	69.52042	315.01144	315.02196	-10.52
82	68.53380	320.00964	320.01702	-7.38
83	67.39174	326.00514	326.00823	-3.10
84	66.65707	329.99376	329.98713	6.62

Order of Fit = 7 RMS error of fit = 5.84 mK
Largest absolute error = 16.61 mK at data point no. 80

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INTERPOLATION TABLE

Calibration Report: 778302 | Sales Order: 90141-1
Sensor Model: CX-1050-SD-HT-1.4L | Serial Number: X93303
Sensor Type: Cernox Resistor | Temperature Range: 1.40K to 325K

Temp (K)	Res. (Ω)	dR/dT (Ω/K)	dlogR/dlogT	Temp (K)	Res. (Ω)	dR/dT (Ω/K)	dlogR/dlogT
1.400	15904.8	-25488.	-2.2436	15.50	863.111	-41.999	-0.75422
1.500	13701.0	-19055.	-2.0861	16.00	842.722	-39.600	-0.75185
1.600	12018.4	-14903.	-1.9840	16.50	823.472	-37.437	-0.75012
1.700	10677.9	-12042.	-1.9172	17.00	805.255	-35.461	-0.74862
1.800	9586.74	-9880.7	-1.8552	17.50	787.983	-33.654	-0.74741
1.900	8685.16	-8219.1	-1.7981	18.00	771.576	-31.998	-0.74648
2.000	7930.94	-6917.8	-1.7445	18.50	755.962	-30.478	-0.74587
2.100	7292.66	-5884.8	-1.6946	19.00	741.078	-29.076	-0.74547
2.200	6747.38	-5050.3	-1.6466	19.50	726.867	-27.781	-0.74530
2.300	6277.46	-4370.8	-1.6014	20.00	713.281	-26.581	-0.74530
2.400	5869.17	-3813.3	-1.5593	21.00	687.802	-24.428	-0.74583
2.500	5511.61	-3351.8	-1.5203	22.00	664.332	-22.554	-0.74688
2.600	5196.28	-2966.2	-1.4842	23.00	642.618	-20.909	-0.74837
2.700	4916.33	-2641.7	-1.4508	24.00	622.450	-19.455	-0.75013
2.800	4666.31	-2366.3	-1.4199	25.00	603.654	-18.162	-0.75217
2.900	4441.74	-2131.0	-1.3913	26.00	586.081	-17.004	-0.75435
3.000	4239.02	-1928.4	-1.3647	27.00	569.607	-15.963	-0.75664
3.100	4055.16	-1752.9	-1.3400	28.00	554.123	-15.021	-0.75903
3.200	3887.69	-1599.9	-1.3169	29.00	539.536	-14.166	-0.76145
3.300	3734.55	-1465.8	-1.2953	30.00	525.764	-13.387	-0.76388
3.400	3594.00	-1347.6	-1.2749	31.00	512.739	-12.675	-0.76632
3.500	3464.56	-1243.0	-1.2557	32.00	500.395	-12.021	-0.76872
3.600	3345.01	-1149.9	-1.2376	33.00	488.679	-11.419	-0.77112
3.700	3234.25	-1066.8	-1.2204	34.00	477.542	-10.864	-0.77347
3.800	3131.36	-992.21	-1.2041	35.00	466.939	-10.349	-0.77572
3.900	3035.55	-925.11	-1.1886	36.00	456.831	-9.8723	-0.77798
4.000	2946.13	-864.47	-1.1737	37.00	447.183	-9.4293	-0.78019
4.200	2784.05	-759.70	-1.1461	38.00	437.962	-9.0162	-0.78229
4.400	2641.07	-672.64	-1.1206	39.00	429.141	-8.6310	-0.78438
4.600	2514.06	-599.57	-1.0970	40.00	420.692	-8.2709	-0.78640
4.800	2400.48	-537.87	-1.0755	42.00	404.818	-7.6169	-0.79026
5.000	2298.32	-485.12	-1.0554	44.00	390.173	-7.0400	-0.79390
5.200	2205.93	-439.82	-1.0368	46.00	376.615	-6.5283	-0.79738
5.400	2121.99	-400.53	-1.0193	48.00	364.023	-6.0718	-0.80063
5.600	2045.37	-366.45	-1.0033	50.00	352.295	-5.6634	-0.80378
5.800	1975.14	-336.50	-0.98812	52.00	341.343	-5.2957	-0.80675
6.000	1910.54	-310.05	-0.97370	54.00	331.089	-4.9636	-0.80955
6.500	1769.54	-256.57	-0.94244	56.00	321.467	-4.6633	-0.81235
7.000	1651.85	-216.01	-0.91536	58.00	312.418	-4.3900	-0.81499
7.500	1551.98	-184.69	-0.89251	60.00	303.891	-4.1409	-0.81758
8.000	1466.09	-159.87	-0.87235	65.00	284.570	-3.6068	-0.82384
8.500	1391.28	-140.03	-0.85552	70.00	267.657	-3.1728	-0.82979
9.000	1325.46	-123.77	-0.84043	75.00	252.711	-2.8173	-0.83611
9.500	1267.02	-110.38	-0.82760	77.35	246.264	-2.6716	-0.83913
10.00	1214.73	-99.133	-0.81609	80.00	239.388	-2.5193	-0.84190
10.50	1167.59	-89.655	-0.80626	85.00	227.443	-2.2663	-0.84697
11.00	1124.85	-81.534	-0.79733	90.00	216.661	-2.0522	-0.85246
11.50	1085.87	-74.561	-0.78965	95.00	206.872	-1.8681	-0.85786
12.00	1050.13	-68.514	-0.78292	100.0	197.939	-1.7089	-0.86335
12.50	1017.22	-63.239	-0.77711	105.0	189.749	-1.5699	-0.86874
13.00	986.802	-58.523	-0.77097	110.0	182.211	-1.4479	-0.87407
13.50	958.612	-54.334	-0.76518	115.0	175.247	-1.3400	-0.87930
14.00	932.372	-50.718	-0.76156	120.0	168.792	-1.2439	-0.88434
14.50	907.814	-47.566	-0.75974	125.0	162.791	-1.1580	-0.88920
15.00	884.766	-44.664	-0.75721	130.0	157.197	-1.0808	-0.89382

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INTERPOLATION TABLE

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Temp (K)	Res. (Ω)	dR/dT (Ω/K)	dlogR/dlogT	Temp (K)	Res. (Ω)	dR/dT (Ω/K)	dlogR/dlogT
135.0	151.970	-1.0111	-0.89821	235.0	91.1573	-0.36206	-0.93338
140.0	147.075	-0.94795	-0.90235	240.0	89.3839	-0.34744	-0.93289
145.0	142.481	-0.89048	-0.90623	245.0	87.6815	-0.33363	-0.93224
150.0	138.162	-0.83803	-0.90983	250.0	86.0463	-0.32059	-0.93143
155.0	134.093	-0.79000	-0.91318	255.0	84.4745	-0.30824	-0.93047
160.0	130.255	-0.74591	-0.91625	260.0	82.9628	-0.29655	-0.92937
165.0	126.628	-0.70532	-0.91905	265.0	81.5080	-0.28547	-0.92814
170.0	123.197	-0.66787	-0.92160	270.0	80.1071	-0.27497	-0.92677
175.0	119.945	-0.63323	-0.92388	273.15	79.2510	-0.26862	-0.92585
180.0	116.860	-0.60112	-0.92591	275.0	78.7574	-0.26499	-0.92528
185.0	113.930	-0.57131	-0.92770	280.0	77.4564	-0.25552	-0.92368
190.0	111.144	-0.54357	-0.92924	285.0	76.2015	-0.24651	-0.92197
195.0	108.491	-0.51772	-0.93055	290.0	74.9905	-0.23794	-0.92015
200.0	105.964	-0.49359	-0.93162	295.0	73.8214	-0.22978	-0.91824
205.0	103.553	-0.47103	-0.93248	300.0	72.6921	-0.22201	-0.91623
210.0	101.251	-0.44990	-0.93312	305.0	71.6007	-0.21460	-0.91414
215.0	99.0514	-0.43010	-0.93356	310.0	70.5455	-0.20753	-0.91197
220.0	96.9479	-0.41150	-0.93380	315.0	69.5248	-0.20079	-0.90973
225.0	94.9345	-0.39402	-0.93384	320.0	68.5371	-0.19435	-0.90741
230.0	93.0060	-0.37756	-0.93370	325.0	67.5809	-0.18819	-0.90503

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F010-04-00_B 06/21/2011

This sensor was tested for repeatability through rapid thermal cycles from room temperature into liquid helium. During this test, the following four lead resistance values were recorded:

Approximately 295 K:	70.5 Ω
Liquid Nitrogen:	247 Ω
Liquid Helium:	2781 Ω

The nitrogen and helium values were recorded in OPEN dewars, so precision comparisons with calibration values or other thermal cycle test values should not be made.

Recommended Operating Parameters:

For sensors calibrated by LSCI, the current to the sensor is adjusted to maintain the sensor output voltage or power at the values listed on the Test Data page.

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BREAKPOINTS 340 FORMAT

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Name: CX-1050-SD-HT-1.4L
Serial number: X93303
Format: 4 ;Log Ohms/Kelvin
Limit: 325.0
Coefficient: 1 ;Negative

Point 1: 1.82981,325.000
Point 2: 1.83715,319.000
Point 3: 1.84402,313.500
Point 4: 1.85102,308.000
Point 5: 1.85818,302.500

Point 6: 1.86548,297.000
Point 7: 1.87293,291.500
Point 8: 1.88055,286.000
Point 9: 1.88833,280.500
Point 10: 1.89628,275.000

Point 11: 1.90440,269.500
Point 12: 1.91271,264.000
Point 13: 1.92120,258.500
Point 14: 1.92989,253.000
Point 15: 1.93878,247.500

Point 16: 1.94788,242.000
Point 17: 1.95720,236.500
Point 18: 1.96673,231.000
Point 19: 1.97561,226.000
Point 20: 1.98468,221.000

Point 21: 1.99396,216.000
Point 22: 2.00346,211.000
Point 23: 2.01317,206.000
Point 24: 2.02312,201.000
Point 25: 2.03331,196.000

Point 26: 2.04375,191.000
Point 27: 2.05445,186.000
Point 28: 2.06542,181.000
Point 29: 2.07667,176.000
Point 30: 2.08823,171.000

Point 31: 2.09890,166.500
Point 32: 2.10983,162.000
Point 33: 2.12104,157.500
Point 34: 2.13253,153.000
Point 35: 2.14433,148.500

Point 36: 2.15645,144.000
Point 37: 2.16891,139.500
Point 38: 2.18173,135.000
Point 39: 2.19492,130.500
Point 40: 2.20700,126.500

Point 41: 2.21939,122.500
Point 42: 2.23215,118.500
Point 43: 2.24528,114.500
Point 44: 2.25882,110.500
Point 45: 2.27279,106.500

Point 46: 2.28722,102.500
Point 47: 2.29840, 99.500
Point 48: 2.30791, 97.000
Point 49: 2.31765, 94.500
Point 50: 2.32762, 92.000

Point 51: 2.33783, 89.500
Point 52: 2.34829, 87.000
Point 53: 2.35902, 84.500
Point 54: 2.37004, 82.000
Point 55: 2.38137, 79.500

Point 56: 2.39303, 77.000
Point 57: 2.40503, 74.500
Point 58: 2.41490, 72.500
Point 59: 2.42500, 70.500
Point 60: 2.43536, 68.500

Point 61: 2.44600, 66.500
Point 62: 2.45693, 64.500
Point 63: 2.46817, 62.500
Point 64: 2.47975, 60.500
Point 65: 2.49107, 58.600

Point 66: 2.50334, 56.600
Point 67: 2.51603, 54.600
Point 68: 2.52781, 52.800
Point 69: 2.53996, 51.000
Point 70: 2.55250, 49.200

Point 71: 2.56547, 47.400
Point 72: 2.57889, 45.600
Point 73: 2.59279, 43.800
Point 74: 2.60560, 42.200
Point 75: 2.61884, 40.600

Point 76: 2.63256, 39.000
Point 77: 2.64680, 37.400
Point 78: 2.66159, 35.800
Point 79: 2.67602, 34.300
Point 80: 2.69101, 32.800

Point 81: 2.70664, 31.300
Point 82: 2.72185, 29.900
Point 83: 2.73771, 28.500
Point 84: 2.75430, 27.100
Point 85: 2.77043, 25.800

Point 86: 2.78732, 24.500
Point 87: 2.80368, 23.300
Point 88: 2.82084, 22.100
Point 89: 2.83893, 20.900
Point 90: 2.85163, 20.100

Point 91: 2.86226, 19.450
Point 92: 2.87326, 18.800
Point 93: 2.88466, 18.150
Point 94: 2.89648, 17.500
Point 95: 2.90782, 16.900

Point 96: 2.91959, 16.300
Point 97: 2.93183, 15.700
Point 98: 2.94352, 15.150
Point 99: 2.95569, 14.600
Point 100: 2.96837, 14.050

Point 101: 2.98160, 13.500
Point 102: 2.99419, 13.000
Point 103: 3.00737, 12.500
Point 104: 3.02119, 12.000
Point 105: 3.03572, 11.500

Point 106: 3.04948, 11.050
Point 107: 3.06392, 10.600
Point 108: 3.07915, 10.150
Point 109: 3.09525, 9.700
Point 110: 3.11040, 9.300

Point 111: 3.12638, 8.900
Point 112: 3.14334, 8.500
Point 113: 3.16138, 8.100
Point 114: 3.18064, 7.700
Point 115: 3.19867, 7.350

Point 116: 3.21787, 7.000
Point 117: 3.23846, 6.650
Point 118: 3.26061, 6.300
Point 119: 3.28529, 5.940
Point 120: 3.30755, 5.640

Point 121: 3.33155, 5.340
Point 122: 3.35759, 5.040
Point 123: 3.38404, 4.760
Point 124: 3.41284, 4.480
Point 125: 3.44212, 4.220

Point 126: 3.46661, 4.020
Point 127: 3.48880, 3.850
Point 128: 3.51252, 3.680
Point 129: 3.53795, 3.510
Point 130: 3.56367, 3.350

Point 131: 3.59131, 3.190
Point 132: 3.62119, 3.030
Point 133: 3.65156, 2.880
Point 134: 3.68223, 2.740
Point 135: 3.71546, 2.600

Point 136: 3.75172, 2.460
Point 137: 3.78856, 2.330
Point 138: 3.82883, 2.200
Point 139: 3.87317, 2.070
Point 140: 3.91833, 1.950

Point 141: 3.96800, 1.830
Point 142: 4.02314, 1.710
Point 143: 4.07941, 1.600
Point 144: 4.13623, 1.500
Point 145: 4.19406, 1.410

Point 146: 4.20153, 1.400

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BREAKPOINTS 91C/93C/330 FORMAT

Calibration Report: 778302 Sales Order: 90141-1
Sensor Model: CX-1050-SD-HT-1.4L Serial Number: X93303
Sensor Type: Cernox Resistor Temperature Range: 1.40K to 325K

Interpolation Method: Lagrangian
Limit: 325.0 (Kelvin)
Format: 4 (Log Ohms/Kelvin)
Number of Breakpoints: 53

No.	Units	Temperature (K)	No.	Units	Temperature (K)
1	1.82982	325.0	31	3.00473	12.6
2	1.83104	324.0	32	3.06070	10.7
3	1.83964	317.0	33	3.11437	9.2
4	1.85884	302.0	34	3.17094	7.9
5	1.87916	287.0	35	3.22371	6.9
6	1.90070	272.0	36	3.28116	6.0
7	1.92357	257.0	37	3.33505	5.3
8	1.94790	242.0	38	3.39018	4.7
9	1.97384	227.0	39	3.44468	4.2
10	2.00156	212.0	40	3.49573	3.8
11	2.03127	197.0	41	3.55558	3.4
12	2.06322	182.0	42	3.60801	3.1
13	2.09772	167.0	43	3.64755	2.9
14	2.13515	152.0	44	3.69164	2.7
15	2.17602	137.0	45	3.74128	2.5
16	2.22100	122.0	46	3.79778	2.3
17	2.27105	107.0	47	3.82914	2.2
18	2.32763	92.0	48	3.89932	2.0
19	2.39305	77.0	49	3.93878	1.9
20	2.46253	63.5	50	3.98167	1.8
21	2.50400	56.5	51	4.02849	1.7
22	2.54691	50.0	52	4.13675	1.5
23	2.58350	45.0	53	4.20153	1.4
24	2.62396	40.0			
25	2.66926	35.0			
26	2.72079	30.0			
27	2.78079	25.0			
28	2.83592	21.1			
29	2.89283	17.7			
30	2.94903	14.9			

Temperature for Resistance Decades:

Res. (Ohms)	Temp. (K)
100	212.817
1000	12.778
10000	1.760

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BREAKPOINTS 234 FORMAT

Calibration Report: 778302
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 90141-1
Serial Number: X93303
Temperature Range: 1.40K to 325K

Maximum Temperature Error:	
1.4 - 10K:	0.012K
10 - 20K:	0.019K
20 - 40K:	0.009K
40 - 100K:	0.019K
> 100K:	0.067K

BP #	Temp. (K)	Res. (Ω)	Log10 Res.	BP #	Temp. (K)	Res. (Ω)	Log10 Res.
1	316.709	69.18310	1.840	46	28.308	549.5409	2.740
2	301.124	72.44360	1.860	47	26.639	575.4399	2.760
3	286.404	75.85776	1.880	48	25.060	602.5596	2.780
4	272.473	79.43282	1.900	49	23.569	630.9573	2.800
5	259.282	83.17638	1.920	50	22.163	660.6934	2.820
6	246.765	87.09636	1.940	51	20.836	691.8310	2.840
7	234.879	91.20108	1.960	52	19.588	724.4360	2.860
8	223.576	95.49926	1.980	53	18.415	758.5776	2.880
9	212.816	100.0000	2.000	54	17.313	794.3282	2.900
10	202.564	104.7129	2.020	55	16.281	831.7638	2.920
11	192.788	109.6478	2.040	56	15.315	870.9636	2.940
12	183.463	114.8154	2.060	57	14.412	912.0108	2.960
13	174.557	120.2264	2.080	58	13.567	954.9926	2.980
14	166.050	125.8925	2.100	59	12.778	1000.000	3.000
15	157.920	131.8257	2.120	60	11.359	1096.478	3.040
16	150.147	138.0384	2.140	61	10.127	1202.264	3.080
17	142.715	144.5440	2.160	62	9.059	1318.257	3.120
18	135.610	151.3561	2.180	63	8.131	1445.440	3.160
19	128.814	158.4893	2.200	64	7.327	1584.893	3.200
20	122.315	165.9587	2.220	65	6.627	1737.801	3.240
21	116.103	173.7801	2.240	66	6.016	1905.461	3.280
22	110.167	181.9701	2.260	67	5.483	2089.296	3.320
23	104.494	190.5461	2.280	68	5.015	2290.868	3.360
24	99.079	199.5262	2.300	69	4.604	2511.886	3.400
25	93.910	208.9296	2.320	70	4.240	2754.229	3.440
26	88.978	218.7762	2.340	71	3.917	3019.952	3.480
27	84.281	229.0868	2.360	72	3.630	3311.311	3.520
28	79.804	239.8833	2.380	73	3.373	3630.781	3.560
29	75.544	251.1886	2.400	74	3.143	3981.072	3.600
30	71.486	263.0268	2.420	75	2.937	4365.158	3.640
31	67.624	275.4229	2.440	76	2.751	4786.301	3.680
32	63.952	288.4032	2.460	77	2.583	5248.075	3.720
33	60.461	301.9952	2.480	78	2.431	5754.399	3.760
34	57.143	316.2278	2.500	79	2.293	6309.573	3.800
35	53.991	331.1311	2.520	80	2.167	6918.310	3.840
36	50.998	346.7369	2.540	81	2.052	7585.776	3.880
37	48.156	363.0781	2.560	82	1.946	8317.638	3.920
38	45.457	380.1894	2.580	83	1.849	9120.108	3.960
39	42.897	398.1072	2.600	84	1.760	10000.00	4.000
40	40.468	416.8694	2.620	85	1.564	12589.25	4.100
41	38.160	436.5158	2.640	86	1.402	15848.93	4.200
42	35.974	457.0882	2.660	87	1.269	19952.62	4.300
43	33.901	478.6301	2.680				
44	31.934	501.1872	2.700				
45	30.071	524.8075	2.720				

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